

Advanced (Habanero)

- Q1.** What is the effect of $y = f(x) + 3$ on the graph of $y = f(x)$?
(A) Shift down by 3
(B) Shift up by 3
(C) Shift left by 3
(D) Shift right by 3
(E) No shift
- Q2.** The graph of $y = -f(x)$ is a reflection of $y = f(x)$ over the
(A) x -axis
(B) y -axis
(C) origin
(D) none
(E) 45° line
- Q3.** What is the effect of $y = f(2x)$ on the graph of $y = f(x)$?
(A) Horizontal stretch
(B) Horizontal compression
(C) Vertical stretch
(D) Vertical compression
(E) No effect
- Q4.** The graph of $y = f(x - 2)$ is a shift of $y = f(x)$ by
(A) 2 units left
(B) 2 units right
(C) 2 units up
(D) 2 units down
(E) no shift
- Q5.** What is the effect of $y = 2f(x)$ on the graph of $y = f(x)$?
(A) Vertical compression
(B) Vertical stretch
(C) Horizontal stretch
(D) Horizontal compression
(E) No effect
- Q6.** The graph of $y = f(-x)$ is a reflection of $y = f(x)$ over the
(A) x -axis
(B) y -axis
(C) origin
(D) none
(E) 45° line
- Q7.** Which of the following is the graph of $y = |f(x)|$?
(A) Reflection over x -axis
(B) Reflection over y -axis
(C) Vertical stretch
(D) Vertical compression
(E) All of the above
- Q8.** What transformation occurs when $y = f(x)$ is changed to $y = f(x) + 2 - 3x$?
(A) Shift up by 2 and left by 3
(B) Shift down by 2 and right by 3
(C) Shift up by 2 and right by 3
(D) Shift down by 2 and left by 3
(E) None

Open Ended Questions (FRQ)

- Q9.** Describe the transformations applied to $y = f(x)$ to obtain $y = -2f(3x - 2) + 1$. Support your answer with 2 – 3 key points.
- Q10.** Compare the graphs of $y = f(x)$ and $y = 2f(x) - 1$. Describe 2 key similarities and 2 key differences.

Chef's Tips

- Provide clear and concise instructions
- Ensure students have necessary materials
- Encourage students to ask questions if unsure